

RLMREC + SASREC

Bridging language-model semantics and sequential recommendation through contrastive alignment

Two small MLP heads on top of vanilla SASRec, trained with InfoNCE against a frozen sentence encoder, lift NDCG@10 by **+60%** on sparse, text-rich data.

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3 datasets · 5 seeds · 14 ablation variants

github.com/DavWexler/RLMRec-SASRec

Two recent ideas pull in opposite directions.

One understands *order*. The other understands *meaning*. Neither does both.

■ SASRec · Kang & McAuley, 2018

Sequential recommenders

Causal self-attention over a user's ordered history of item IDs.

-
- + Captures temporal dynamics — what you watch *next*.
 - Every item is an opaque integer. Title, genre, description: invisible.

■ RLMRec · Ren et al., WWW 2024

LLM-aligned recommenders

Contrastively align user/item embeddings with frozen sentence-transformer text.

-
- + Embeddings reflect topical meaning, not just co-occurrence.
 - Only ever tested on graph CF backbones — no notion of sequence.

Can one model understand **order** and **meaning** at the same time?

The published RLMRec paper benchmarks six graph-CF backbones — GCCF, LightGCN, SGL, SimGCL, DCCF, AutoCF. **Every one ignores interaction order by design.** Whether the same auxiliary alignment helps a sequential transformer was left open.

Three concrete contributions.

01

A drop-in extension of SASRec

Two two-layer MLP projection heads map SASRec's 50-d state into BGE-large's 1024-d semantic space. Trained with symmetric InfoNCE against frozen profile embeddings. *Backbone, prediction head and BCE loss are all standard.*

02

A fair three-way head-to-head

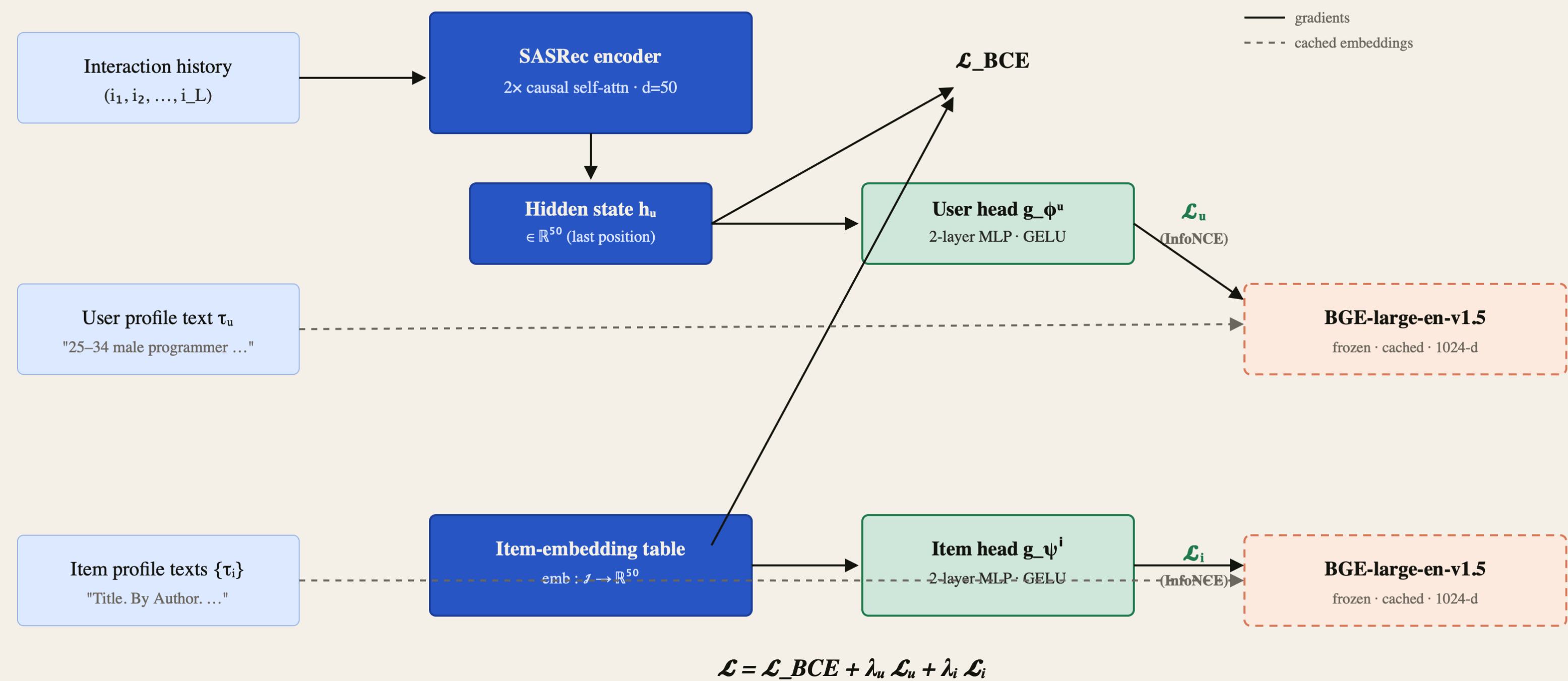
SASRec, RLMRec+LightGCN, and RLMRec+SASRec on three datasets that span the density spectrum. **Same splits, same evaluator, same five seeds** for every (model, dataset) cell. Full-rank leave-one-out — no sampled negatives.

03

A 14-variant ablation grid

Each row flips *one* knob: loss components, λ sweep, τ sweep, projection-head architecture, sentence encoder. Paired t-tests against an in-ablation reference at held-constant seeds and epoch budget.

Two projection heads, one frozen text encoder.



Architecture (Figure 1)

The training objective is one weighted sum.

$$\mathcal{L} = \mathcal{L}_{BCE} + \lambda_u \mathcal{L}_u + \lambda_i \mathcal{L}_i$$

$\mathcal{L}_{BCE}$

Sequential CF loss

Masked BCE on SASRec hidden states against positives and sampled negatives.

$\mathcal{L}_u$ · symmetric InfoNCE

User alignment

Pulls $g_\varphi^u(h_u)$ toward $f_{enc}(\tau_u)$. In-batch negatives.

$\mathcal{L}_i$ · symmetric InfoNCE

Item alignment

Pulls $g_\psi^i(\text{emb}(i))$ toward $f_{enc}(\tau_i)$. Unique positives in batch, sub-sampled to K=512.

Defaults (RLMRec paper): $\lambda_u = \lambda_i = 0.1$ · $\tau = 0.1$

Three datasets span the density spectrum.

Same splits, same five seeds, same full-rank leave-one-out evaluator for every cell.

DATASET	USERS	ITEMS	~ INTER. / USER	ITEM TEXT	DENSITY
MovieLens-1M Harper & Konstan, 2015	6,040	3,706	~165	Title + 1–3 genres	dense
Yelp Open Dataset 2018-01-01 onward, 5-core	104,072	54,056	~10	Name + categories + city + stars	sparse
Amazon-Books (2023) Hou et al., 2024 · 5-core	156,213	112,362	~6	Title + author + 4-level cat + 400-char description	very sparse · rich text

ML-1M tests whether the recipe breaks where SASRec is already near-optimal. Yelp and Amazon-Books are the natural transfer test — they sit in the regime RLMRec was originally designed for.

The lift tracks data sparsity and item-text richness.

AMAZON-BOOKS

very sparse · rich text

+60.0%

NDCG@10: 0.0222 → 0.0355 HR@10:
+50.6% · MRR: +53.2%

worst RLMRec+SASRec-seed (0.0348) >
best SASRec-seed (0.0230)

YELP

sparse · medium text

+30.8%

NDCG@10: 0.0125 → 0.0164 HR@10:
+28.7% · MRR: +24.4%

RLMRec+LightGCN also beats SASRec —
alignment alone helps.

MOVIELENS-1M

dense · short text

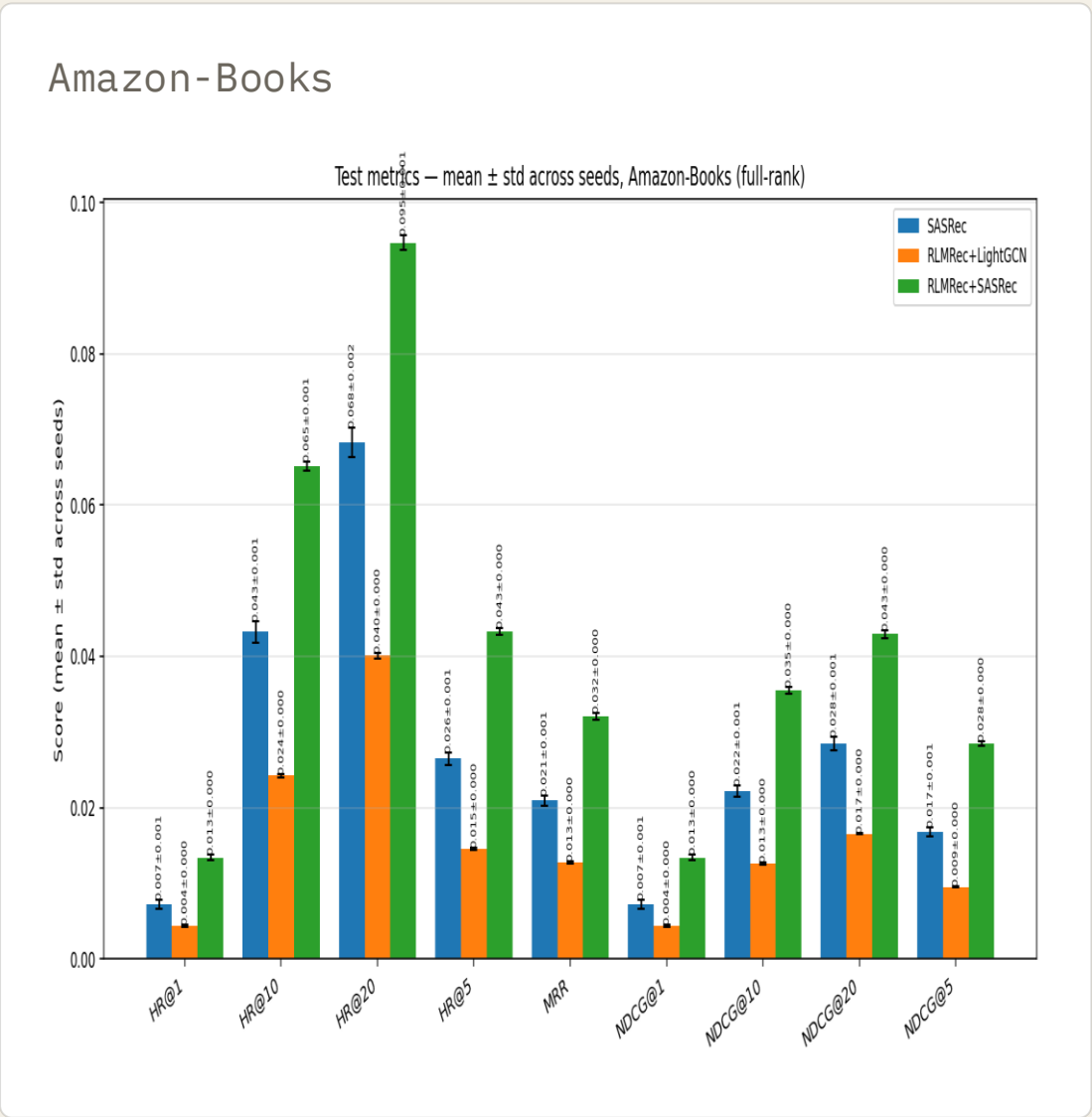
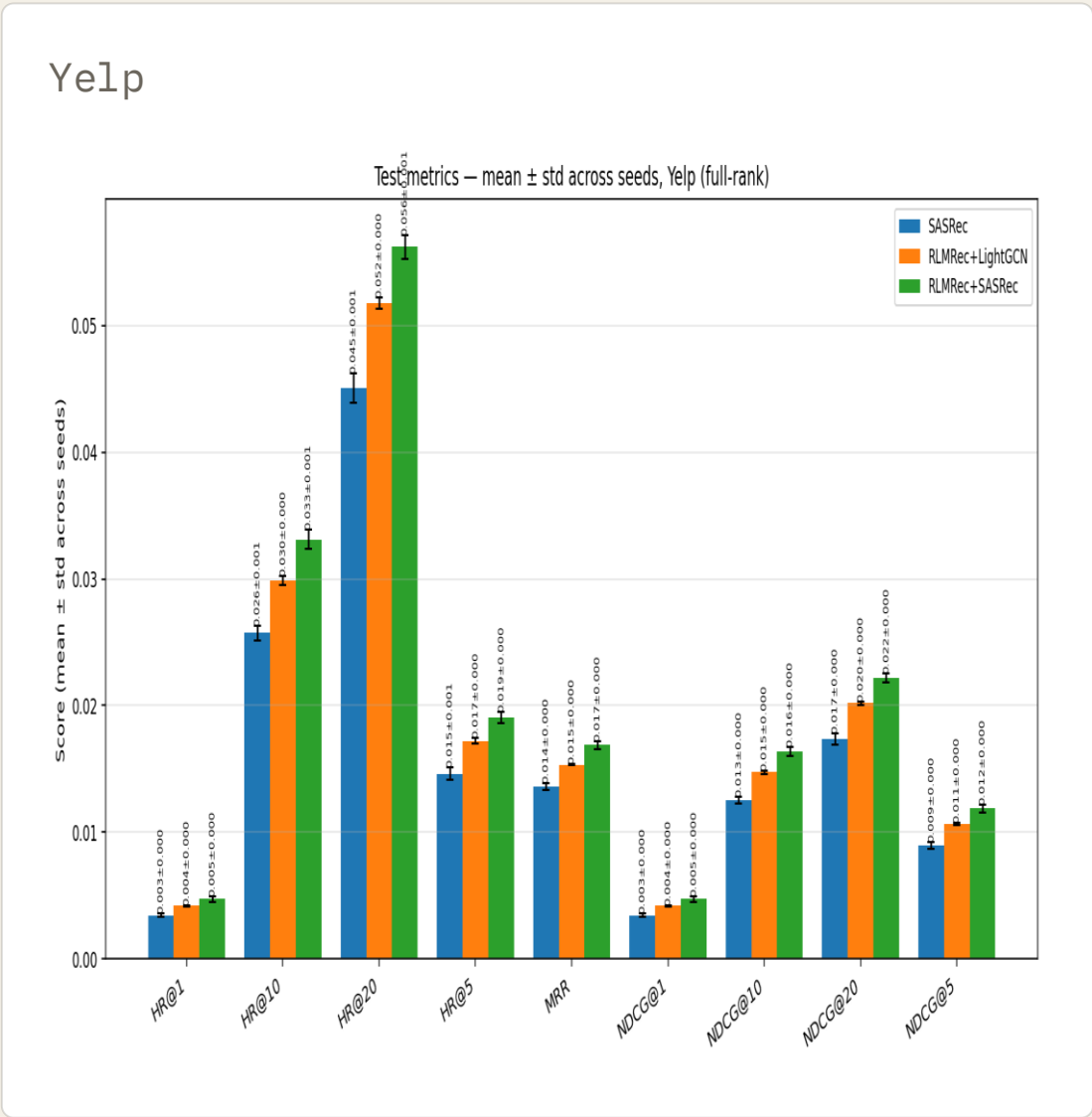
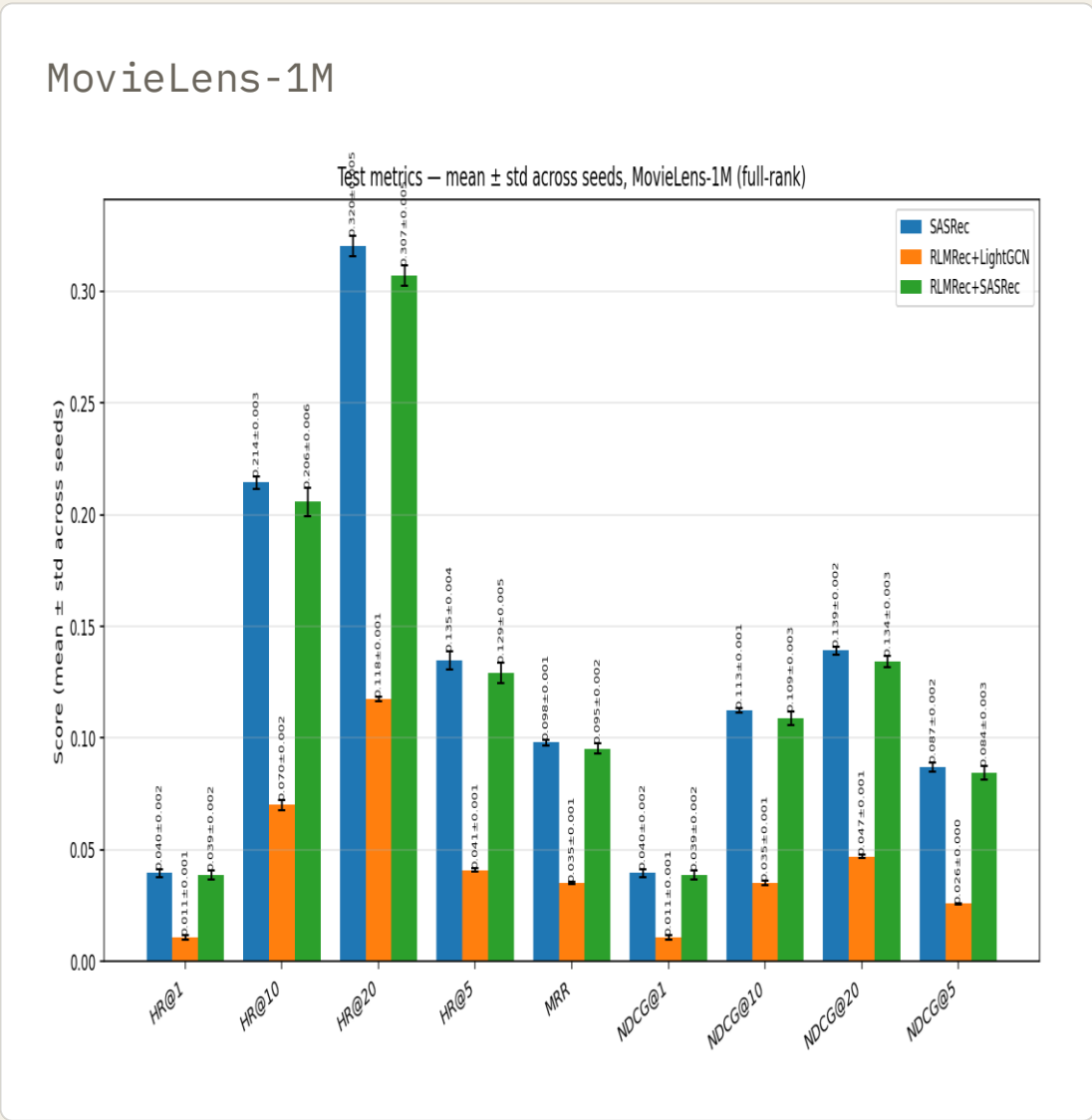
−3.3%

NDCG@10: 0.1126 → 0.1089 Within
1.5σ. Approximately neutral.

Item text = title + 1–3 genres. No
semantic room to add value.

All nine metrics agree on the ordering.

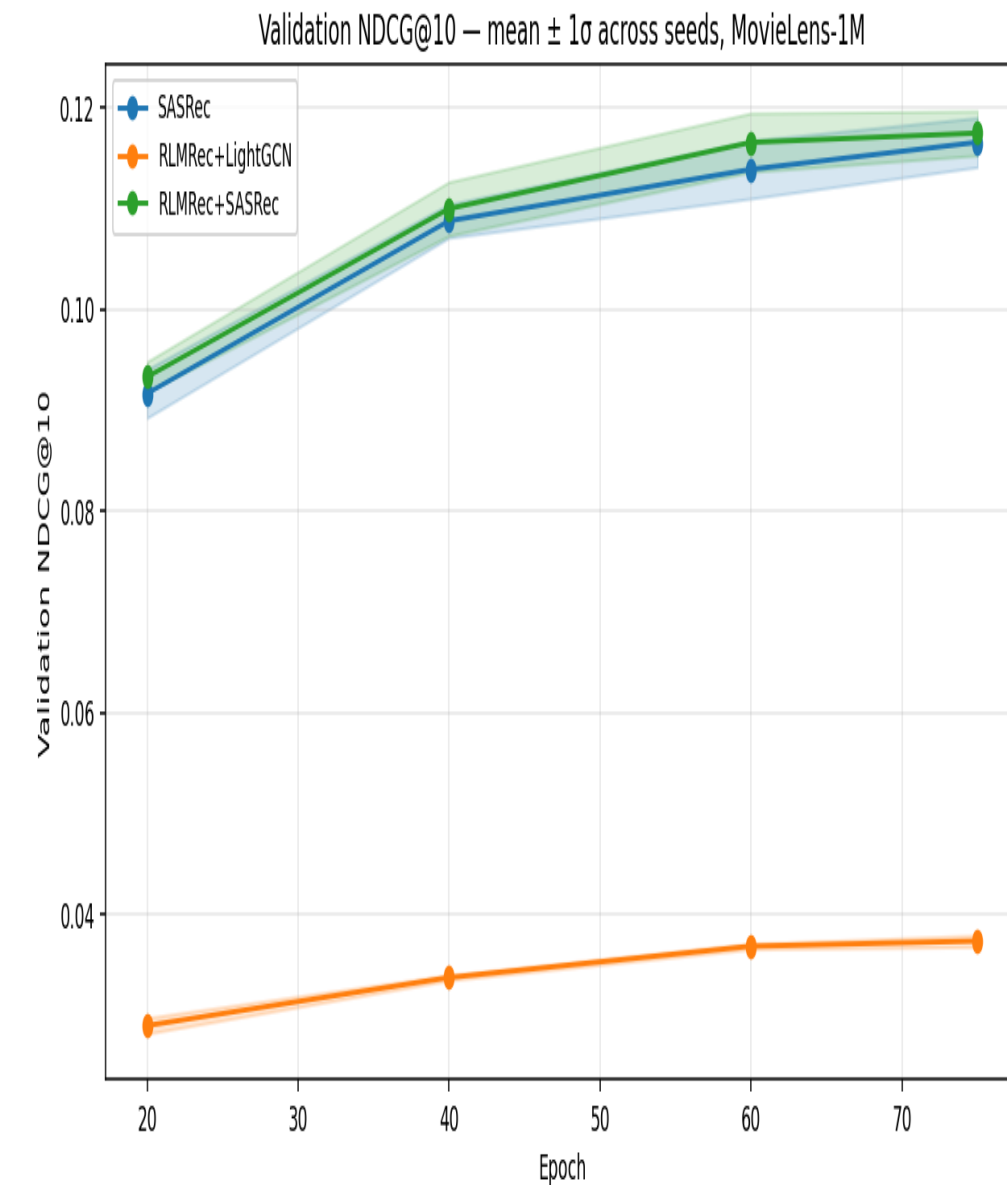
SASRec RLMRec+LightGCN RLMRec+SASRec (*ours*)



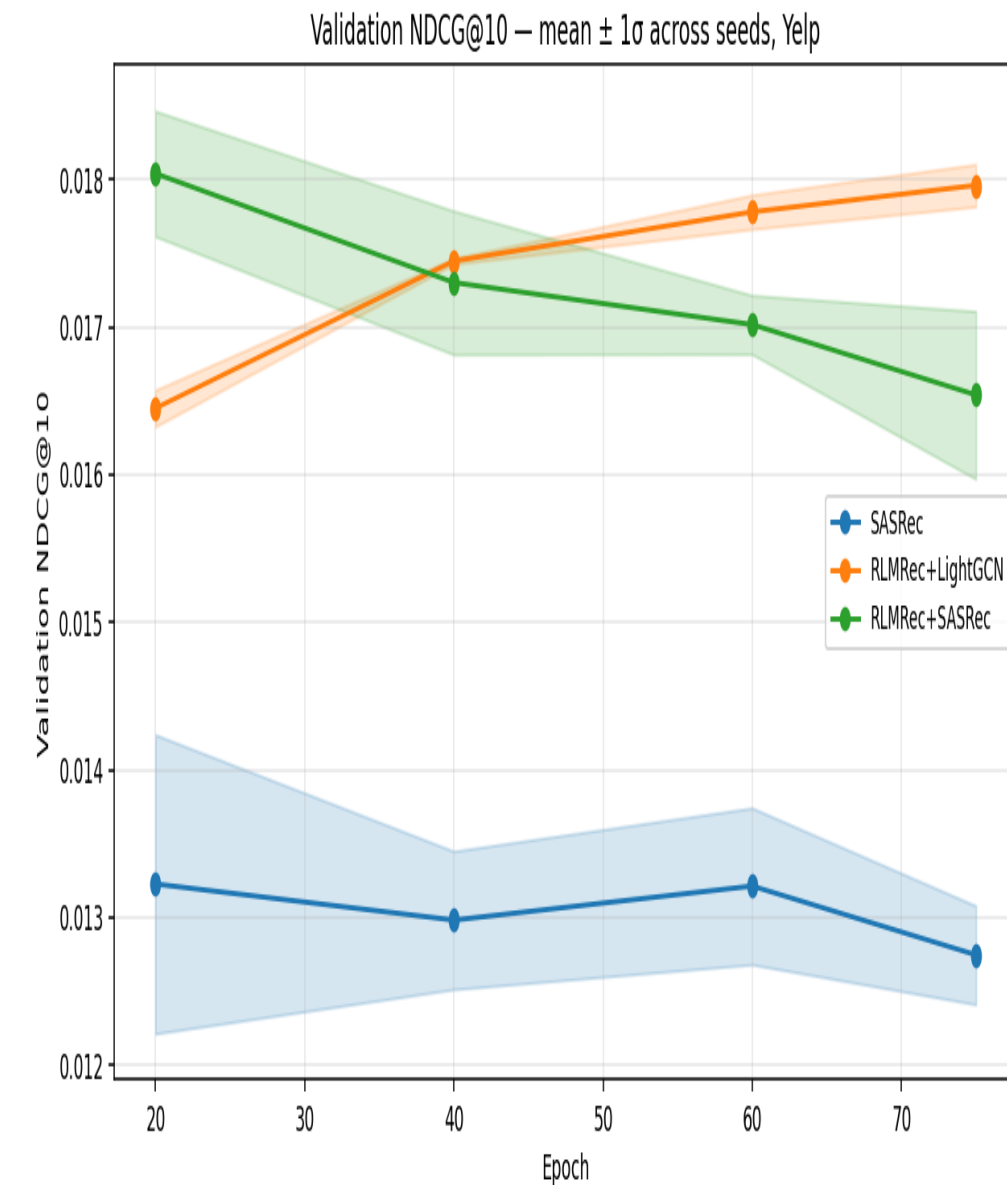
Amazon-Books: green dominates by a wide margin. · Yelp: green wins everywhere; orange also beats blue. · ML-1M: blue and green track; orange trails (LightGCN has no temporal signal).

The gap on text-rich data opens within 20 epochs.

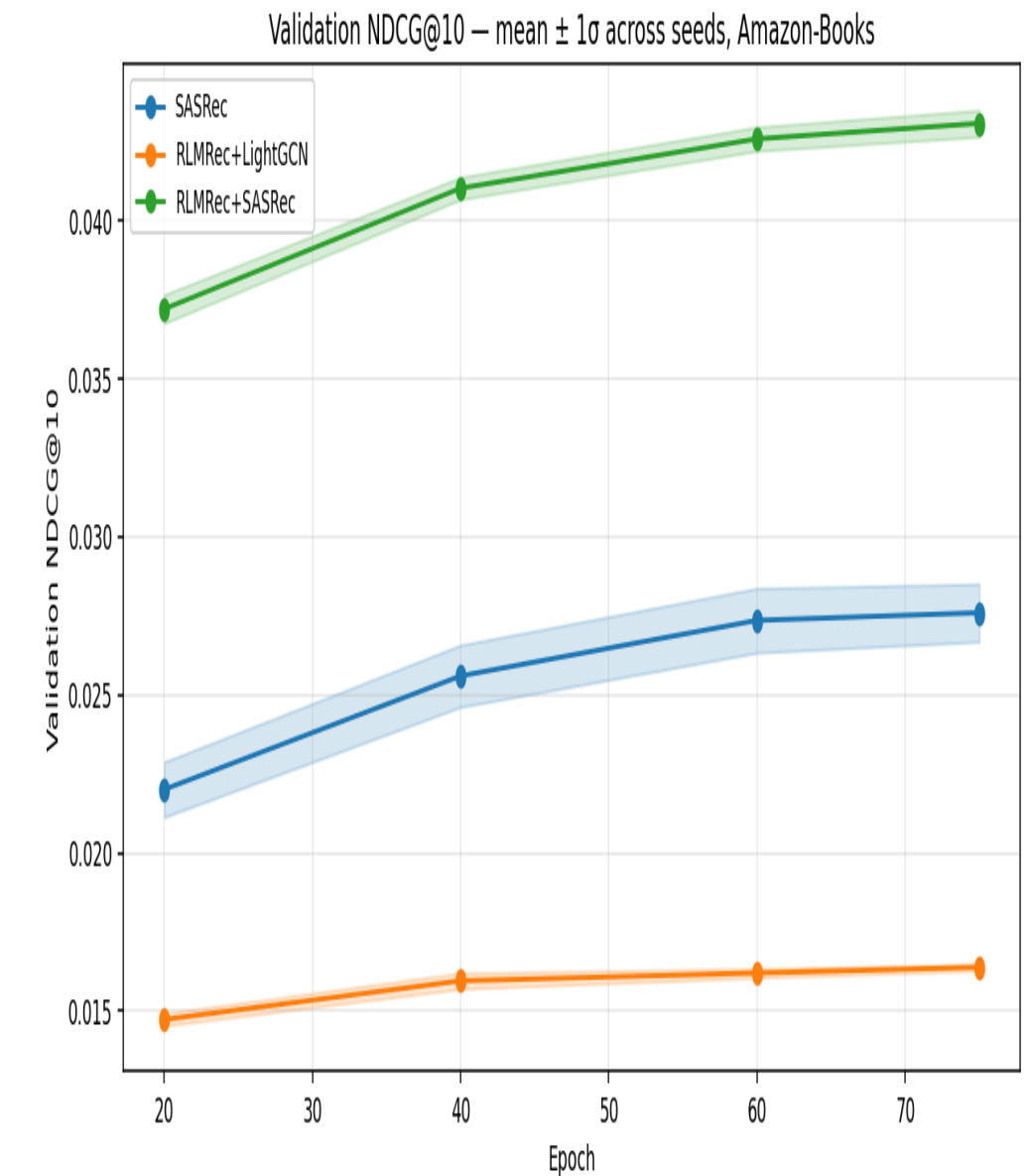
MovieLens-1M · two curves within 1σ



Yelp · gap widens through training



Amazon-Books · separation by epoch 20



Which design choices actually matter?

01 · Loss components

Both alignment heads matter on sparse data, **asymmetrically**.
Amazon-Books: user head dominates (−32% NDCG to remove).
Yelp: item head dominates (−16%).

02 · λ sweep

Bell-shaped on every dataset. Default $\lambda = 0.1$ sits near the peak. $\lambda = 1.0$ collapses by 19–39%.

03 · InfoNCE temperature

The most robust knob. Entire τ sweep stays within $\pm 10\%$ on every dataset.

04 · Projection head

Linear matches MLP on dense / short-text data. MLP wins by 18.6% on Amazon-Books — the nonlinearity decouples CF ranking from semantic alignment.

05 · Sentence encoder

Swapping BGE-large (1024-d) for MiniLM (384-d) costs 20% NDCG on Amazon-Books, $\leq 7\%$ elsewhere. When the text is rich, encoder capacity matters; when it isn't, none of these knobs save you.

Three caveats on the claim.

01

Template profiles, not LLM-authored

RLMRec originally used an LLM to write user/item profiles; I used a deterministic template builder. Reported gains are almost certainly a *lower bound*.

02

Backbone untuned per dataset

A higher dropout or longer max sequence is known to help on Amazon-Books. Defaults kept across all three datasets to keep the comparison clean.

03

Single-step next-item only

Evaluation is leave-one-out next-item — the protocol both SASRec and RLMRec target. Session-level or multi-step retrieval may behave differently.

CONCLUSION

RLMRec-style alignment transfers cleanly to a sequential backbone — in the data regime it was designed for.

+60%

NDCG@10 on Amazon-Books

2 MLPs

total engineering footprint on top of SASRec

~0 ms

added wall-clock at training (cache + tiny logit matrices)

Full code, configs, seeds, and the cached semantic embeddings are released. `python run_everything.py` reproduces Table 1 end to end.

Questions?

§ appendix

Deep dive

Profile construction · backbone in detail · InfoNCE derivation · projection-head design ·
Algorithm 1 · full ablation table · comparison to original papers.

Density and text richness pull in opposite directions.

MovieLens-1M

Harper & Konstan 2015 · ratings ≥2000

users	6,040
items	3,706
~ inter/user	~165
item text	title + genres
user text	demo + top-3 genres + last 5 films

"Star Wars (1977). Genres: Action, Adventure, Sci-Fi."

Yelp Open Dataset

2022 release · reviews ≥ 2018-01-01 · 5-core

users	104,072
items	54,056
~ inter/user	~10
item text	name + categories + city + stars
user text	review_count + top-3 cats + last 5 places

"Joe's Pizza. Categories: Italian, Pizza. Located in NYC, NY. 4.0 stars."

Amazon-Books 2023

Hou et al. 2024 · 5-core · ≥ 2018-01-01

users	156,213
items	112,362
~ inter/user	~6
item text	title + author + 4-level cat + 400-char desc
user text	total reviews + top-3 cats + last 5 titles

"The Pragmatic Programmer. By Hunt & Thomas. Categories: Computers › Software... ...4.5 average stars."

Profiles are templated, not LLM-authored.

Each profile is a single short string built deterministically from structured metadata.

ITEM PROFILE · ML-1M

Star Wars (1977). Genres: Action, Adventure, Sci-Fi.

USER PROFILE · ML-1M

A 25–34 male programmer who enjoys Action, Sci-Fi, Adventure movies. Recently watched: Inception; The Matrix; Star Wars; ...

Short, structured. Little to align with beyond IDs.

ITEM PROFILE · Amazon-Books

The Pragmatic Programmer. By Hunt & Thomas. Categories: Computers › Software › Methodology. Author: Andrew Hunt, David Thomas. A guide for working programmers... 4.5 average stars.

USER PROFILE · Amazon-Books

82 reviews across Computers, Fantasy, Biography. Recently read: Designing Data-Intensive Apps; The Hobbit; Algorithms; ...

Paragraph-scale. The richness is exactly what the alignment latches onto.

SASRec backbone — paper-faithful, unchanged.

Embedding

$$E^{(0)}_t = \forall d \cdot emb(i_t) + pos(t)$$

Masked BCE loss

$$\mathcal{L}_{BCE} = - (1/|M|) \sum_{(u,t) \in M} \log \sigma(H_{u,t}^{\top} emb(i_t^+)) + \log(1 - \sigma(H_{u,t}^{\top} emb(i_t^-)))$$

One sampled negative per position. At inference, only the last-position state is scored against every item, with training + validation items masked to $-\infty$.

Hyperparameters (held constant across datasets)

hidden dim d	50
attn. blocks N	2
attention heads	1
dropout	0.2
batch size	128
max seq. length	200
optimizer	Adam · lr 1e-3
β_1, β_2	0.9, 0.98

Sequences left-padded to 200 with ID 0. Pre-norm causal multi-head attention with point-wise feed-forward (two 1×1 convs + residual). Final LayerNorm before scoring.

Symmetric InfoNCE on L2-normalized projections.

Let $q = g_\theta(x)$ be the projected CF vector and k be the L2-normalized frozen semantic target. Form the cosine-similarity logit matrix

$$M_{ij} = (\hat{q}_i^\top \hat{k}_j) / \tau$$

Minimize the average of two cross-entropies on this matrix:

$$\mathcal{L}_{\text{InfoNCE}} = \frac{1}{2} [CE(M, I) + CE(M^\top, I)]$$

Matched pairs sit on the diagonal. Averaging the two views — "for each query, which key matches?" and "for each key, which query matches?" — keeps the gradient symmetric. The variant used in the RLMRec paper.

User head ($\mathcal{L}_u$)

Anchor = batch's projected last-position states $g_{\varphi^u}(h_u)$. Target = cached $\mathbf{f}_{\text{enc}}(\tau_u)$. In-batch negatives = other users in the batch.

Item head ($\mathcal{L}_i$)

Anchor = unique positives in the batch, sub-sampled to K=512. Same emb-row used by BCE — no separate aligned-only copy.

defaults

$\tau = 0.1 \cdot K = 512$

sweep: $\tau \in \{0.05, 0.2, 0.5\}$

Two-layer MLP. The nonlinearity is the design choice.

$$g_{\theta}(x) = W_2 \cdot \text{GELU}(W_1 x + b_1) + b_2$$

$W_1 \in \mathbb{R}^{1024 \times 50}$

$W_2 \in \mathbb{R}^{1024 \times 1024}$

hidden width $h \quad \max(d, D) = 1024$

CF dim $d \quad 50$ (= SASRec width)

semantic dim $D \quad 1024$ (= BGE-large)

Why nonlinear, not linear?

Following the SimCLR result [Chen et al. 2020] : a nonlinear head decouples the underlying representation from the contrastive objective.

The CF embedding is free to preserve ranking-relevant information that does not perfectly align with the semantic target. A linear projection cannot make that decoupling.

Ablation: linear costs −18.6% NDCG on Amazon-Books ($p < 0.01$); neutral on Yelp / ML-1M. The decoupling only earns its keep when there is rich text to absorb.

Item head reuses the same *emb* -row that the BCE head scores against. The shared row must satisfy both objectives — that sharing is the whole point.

One training epoch of RLMRec+SASRec.

```

01: Shuffle user index  $\pi$ 
02: for  $b \in \{0, \text{batch\_size}, 2 \cdot \text{batch\_size}, \dots\}$  do
03:   Gather  $B = \pi[b : b + \text{batch\_size}]$  users
04:   Build  $(S, P, N)$  – input seqs, positives, sampled negatives
05:    $H \leftarrow \text{SASRec.encode}(S) \# (|B|, L, d)$ 
06:    $\mathcal{L}_{\text{BCE}} \leftarrow \text{Eq. 3 from } H, P, N$ 
07:    $Q_u \leftarrow g_\phi^u(H_{:,L-1,:})$ 
08:    $\mathcal{L}_u \leftarrow \text{InfoNCE}(Q_u, E_u[B], \tau)$ 
09:    $I \leftarrow \text{unique non-padding items in } P, \text{ sub-sampled to } K = 512$ 
10:    $Q_i \leftarrow g_\psi^i(\text{emb}(I))$ 
11:    $\mathcal{L}_i \leftarrow \text{InfoNCE}(Q_i, E_i[I], \tau)$ 
12:    $\mathcal{L} \leftarrow \mathcal{L}_{\text{BCE}} + \lambda_u \cdot \mathcal{L}_u + \lambda_i \cdot \mathcal{L}_i$ 
13:   Adam step on  $\nabla \mathcal{L}$ 
14: end for

```

Both InfoNCE logit matrices are tiny relative to the attention block: $|B| \times |B| = 128 \times 128$ for the user head; $K \times K \leq 512 \times 512$ for the item head. Alignment adds essentially **no wall-clock overhead**.

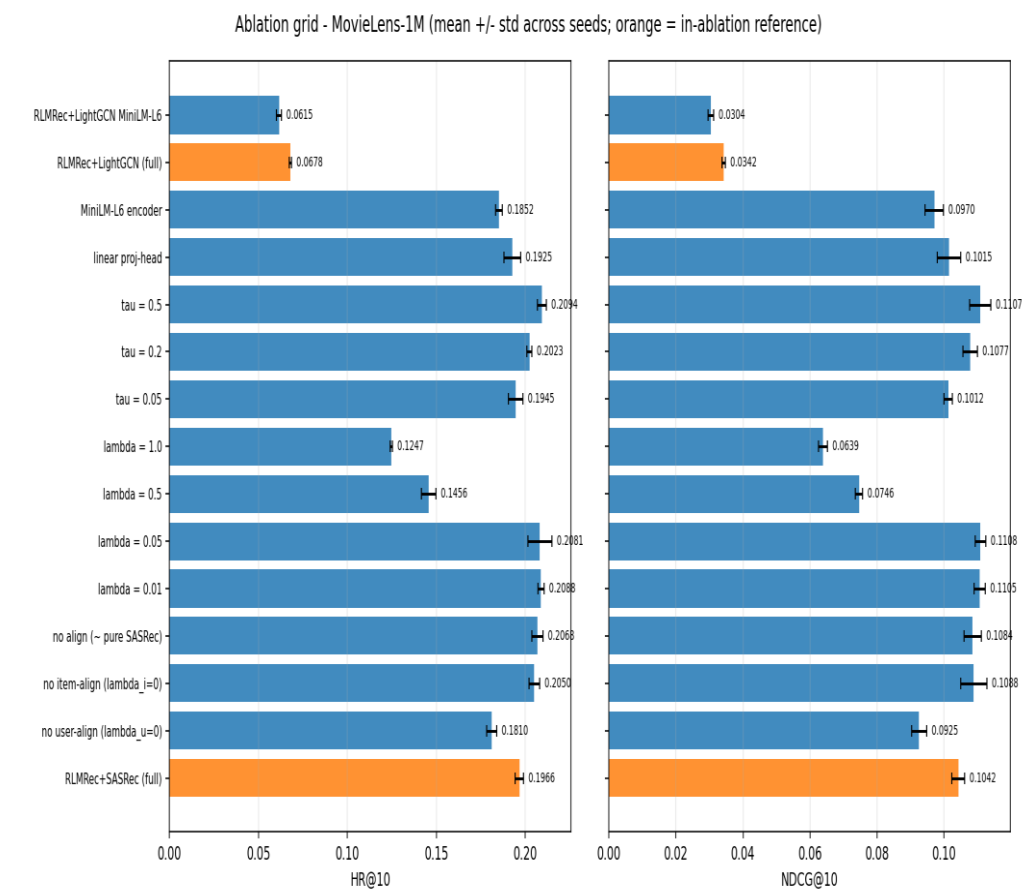
Full ablation grid · NDCG@10 vs. sasrec_full

VARIANT	ML-1M Δ%	YELP Δ%	AMAZON Δ%
LOSS COMPONENTS			
no user-align ($\lambda_u = 0$)	-11.2 *	+2.1	-32.0 **
no item-align ($\lambda_i = 0$)	+4.5	-15.7 *	-8.7 *
no align (~ pure SASRec)	+4.1	-25.3 **	-38.8 **
ALIGNMENT WEIGHT λ SWEEP			
$\lambda = 0.01$	+6.1	-8.9 **	-19.5 **
$\lambda = 0.05$	+6.4 *	+1.7	-1.5
$\lambda = 0.5$	-28.4 **	-16.0 **	-11.0 *
$\lambda = 1.0$	-38.7 ***	-26.0 ***	-19.0 **
INFONCE TEMPERATURE τ · ARCHITECTURE / ENCODER			
$\tau = 0.05$	-2.9	-7.0	-6.6 **
$\tau = 0.2$	+3.4	+3.1	-0.6
$\tau = 0.5$	+6.3	-3.8	-9.5 *
linear proj-head	-2.6	-1.2	-18.6 **
MiniLM-L6-v2 encoder	-6.9	-3.3	-20.1 **

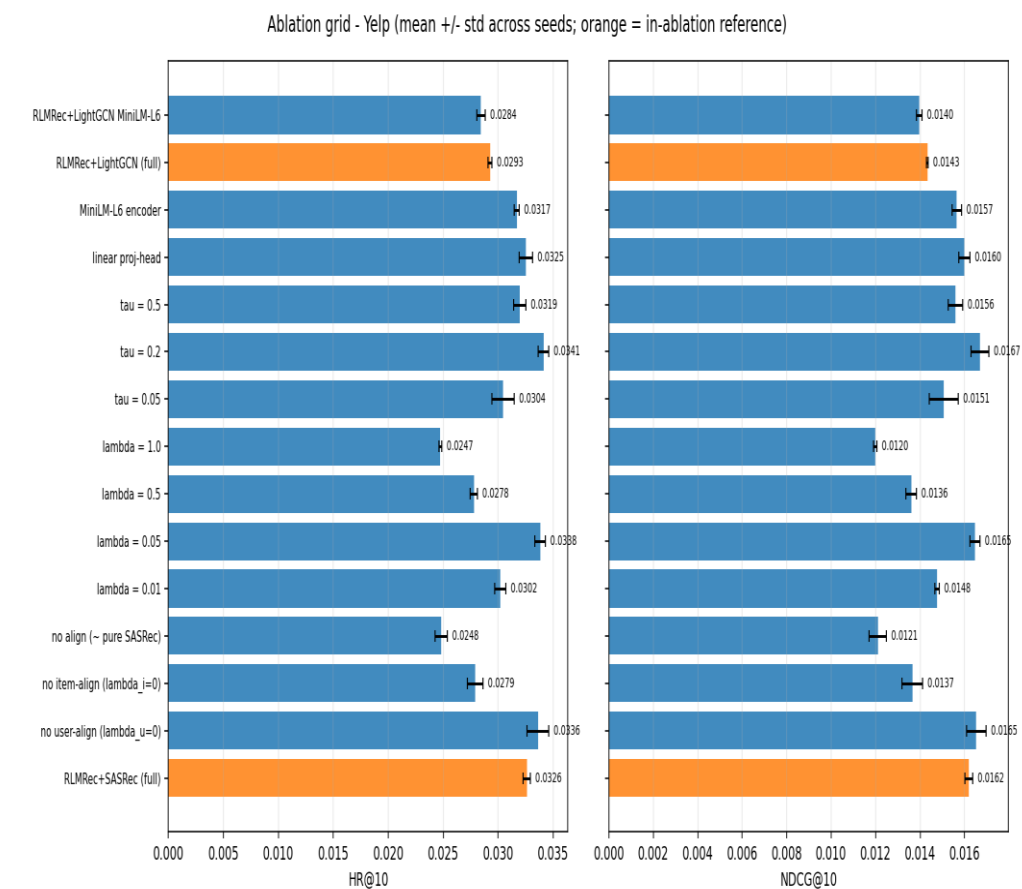
3 seeds × 50 epochs per cell. Paired two-sided t-tests vs. sasrec_full: * p<.05 ** p<.01 *** p<.001.
Ablation table (Table 2)

HR@10 & NDCG@10 per variant • orange = sasrec_full reference.

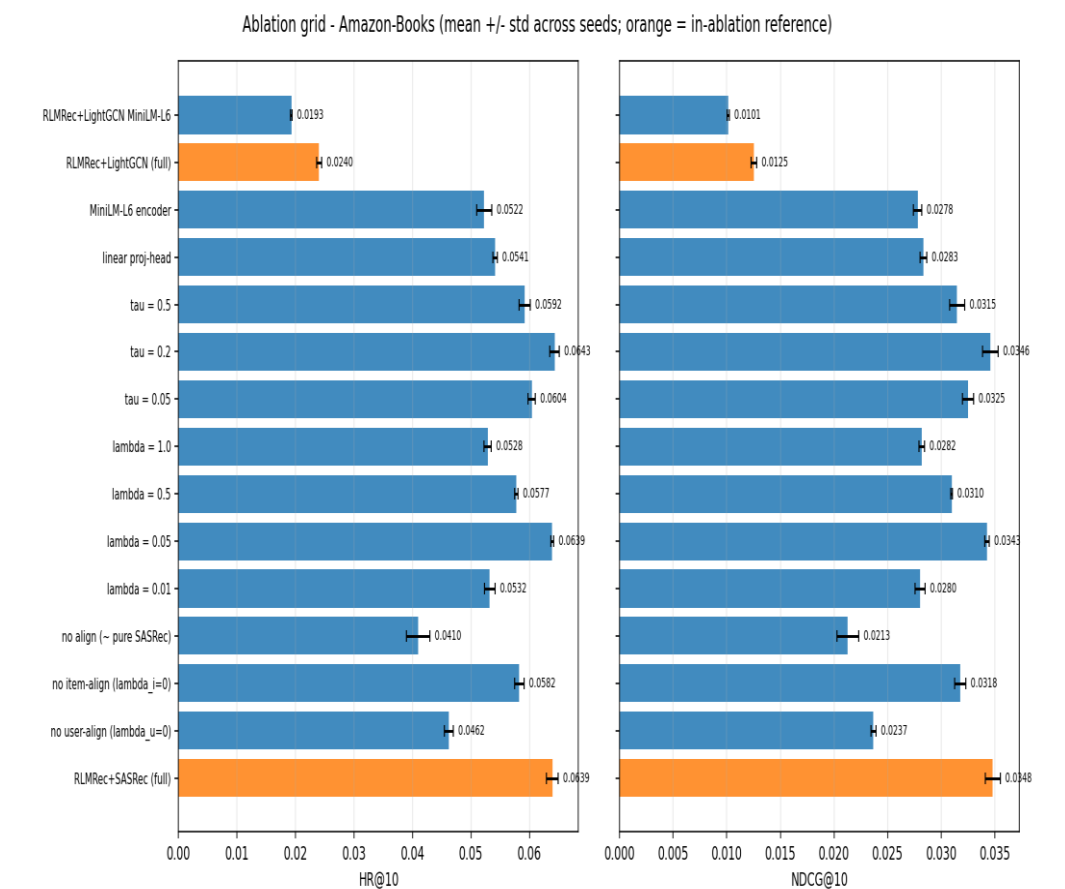
MovieLens-1M



Yelp



Amazon-Books



ML-1M: ref. sits mid-cluster — alignment adds nothing here. • Yelp: ref. dominates — item head removal is the most costly single ablation. • Amazon-Books: ref. clearly best — both heads, the MLP, and the encoder all matter.

Ablation bars

Four natural extensions.

01

LLM-authored profiles

Replace the deterministic template with a Claude / GPT-4 profile builder. Expected: Yelp and Amazon-Books gains grow further; ML-1M could plausibly flip positive once a movie's profile is a paragraph instead of three words.

02

Generative alignment (RLMRec-Gen)

The contrastive head is the simpler of the two objectives in the original paper. Test whether the masked-autoencoder variant carries similarly large gains on the sequential side.

03

Multi-encoder consensus

Two frozen encoders (e.g. one small + one large, or English + multilingual) contributing separate alignment heads — trading encoder dimension against representational diversity.

04

Cold-start via head inversion

An unseen item has no learned ID embedding, but it does have a semantic embedding. The item-projection head could in principle be inverted to bootstrap a synthetic CF embedding for new items at inference time.

THANK YOU

Questions?

paper RLMRec+SASRec: Bridging Language-Model Semantics and Sequential Recommendation

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code github.com/DavWexler/RLMRec-SASRec

reproduce `python run_everything.py · python run_ablations.py`